NSE Cross-Sectional Alpha Research: do classic price signals survive Indian costs and an honest holdout?

Apurv Sonawane

November 10, 2026

Abstract

We ask whether eight classic daily price and volume signals predict cross-sectional returns on the National Stock Exchange of India after realistic costs, using a survivorship-free point-in-time panel built from exchange archives, a pre-registered protocol, a trial ledger and a locked out-of-sample window read once. Development (walk-forward, 2017–2024): long-only top-decile book net information ratio against the Nifty 500 price index not run (gross not run), deflated Sharpe not run with not run logged trials. Holdout (2024-11-01 to 2026-10-30): net IR not run, 95% bootstrap interval [not run, not run]; F&O long-short net Sharpe not run. Every number in this paper is rendered from the run files by `qr ledger`.

1 Introduction

Do classic price signals survive Indian transaction costs and an honest holdout? Three things make this answer checkable. First, a survivorship-free daily NSE panel (not run sessions, not run securities, not run later-delisted names that were in the liquid universe) rebuilt from raw exchange files. Second, a locked protocol: a pre-registration tag before the first IC, an append-only trial ledger, and a holdout read exactly once. Third, a cost model that changes with the tax rules, including the April 2026 rise in stock-futures STT from 0.02% to 0.05%, which falls inside the holdout window.

2 Data

Raw files come from the NSE static archives (cash bhavcopy in the legacy and UDiFF formats, delivery files, F&O bhavcopy, index files), downloaded at no more than one request per second with an honest User-Agent and an on-disk cache. A permanent security id links symbol changes through ISIN.

Corporate actions. NSE's corporate-action export (a manual, cookie-gated download) was not used. Splits and bonuses are inferred from the data (the PRD's fallback, refined): a one-day move above 25% whose price ratio is within 12% of a common split, bonus or consolidation ratio is treated as that action. For stocks without F&O contracts this is safe, because their price bands (at most 20%) rule out a genuine move of that size. F&O stocks have no band, so the ratio must be confirmed by the implied lot size of a futures contract alive on both days (NSE changes the lot of existing contracts only for a corporate action) or by open interest. Unconfirmed F&O moves are genuine and kept (for example SBI +28% on 2017-10-25 and Yes Bank −56% on 2020-03-06), except demerger-like days (open interest up at least 1.8× with the lot unchanged), which are nulled.

Non-F&O moves that match no ratio are nulled and logged. The PRD's named checks pass (IRCTC 2021-10-28, factor 1/5; Infosys 2018-09-04, factor 1/2). The bhavcopy's own previous-close field turned out not to be split-adjusted, so it could not be used. Dividends are not captured: stock returns are *price* returns, and the long-only book is compared with the Nifty 500 *price* index.

Universe. Rebuilt each month from data dated before the month starts: EQ/BE series, at least 252 sessions of history, at least 55 trading days in 63 sessions, close at least Rs 10, median traded value at least Rs 1 crore, top 750 by value. Data-quality gates and survivorship counts are in the ledger (`data.*`).

3 Method

3.1 Signals and evaluation

Eight signals with signs and windows fixed before any IC was computed (`PREREG.md`): MOM_12_1 [Jegadeesh and Titman, 1993], REV_5, LOWVOL_63, IVOL_63 [Ang et al., 2006], AMIHUD_63 [Amihud, 2002], ABVOL_5_63, DELIV_ABN and HIGH52 [George and Hwang, 2004]. Each is winsorized at 1/99, standardized by date, neutralized to sector dummies, a shrunk Dimson beta and size, and re-standardized. We report rank IC at horizons 1, 5, 10 and 21 sessions with Newey–West t statistics [Newey and West, 1987] and Holm adjustment across signals; a signal passes at $t \geq 3$ after Holm correction.

3.2 Combination

Equal weight, ridge and LightGBM are compared on eight expanding-window folds with a 27-session purge and embargo. The headline is the better of ridge and LightGBM only if it beats equal weight with Newey–West $t \geq 2$ on the IC difference, otherwise equal weight [López de Prado, 2018, Gu et al., 2020].

3.3 Portfolios and costs

P1 is a long-only equal-weight book (enter in the top 10%, hold to the top 20%, weekly, signal at the close and fill at the next close, 10% of ADV cap) against the Nifty 500 price index (stock returns exclude dividends). P2 is long the top quintile in cash and short the bottom quintile in near-month stock futures. Costs are per side: dated STT, exchange charge, SEBI fee, stamp duty, brokerage (assumed 3 bp delivery, 1 bp futures), GST and the DP charge, plus an Abdi–Ranaldo half-spread [Abdi and Ranaldo, 2017] and square-root impact $Y\sigma\sqrt{Q/\mathrm{ADV}}$ with $Y = 1$. Table 1 lists the dated statutory and broker rates (generated from `config/costs.yaml`). Capacity is the smaller of the AUM at which the net metric halves and the AUM at which more than 20% of desired trading is blocked by the ADV cap.

3.4 Validation protocol

A pre-registration commit and tag precede the first IC; `oos.lock` makes every loader blind after 2024-10-31; every signal, combiner fold and base portfolio is logged to the trial ledger before results print; the deflated Sharpe ratio [Bailey and López de Prado, 2014] uses the true ledger count; confidence intervals use a stationary block bootstrap [Politis and Romano, 1994].

Component	Segment	Side	From	Rate (bp)
stt	Delivery	both	2012-01-01	12.500
stt	Delivery	both	2013-06-01	10.000
stt	Stock futures	sell	2012-01-01	1.000
stt	Stock futures	sell	2023-04-01	1.250
stt	Stock futures	sell	2024-10-01	2.000
stt	Stock futures	sell	2026-04-01	5.000
exchange	Delivery	both	2012-01-01	0.307
exchange	Stock futures	both	2012-01-01	0.183
sebi	Delivery	both	2012-01-01	0.010
sebi	Stock futures	both	2012-01-01	0.010
stamp	Delivery	buy	2012-01-01	1.500
stamp	Stock futures	buy	2012-01-01	0.200
brokerage	Delivery	both	2012-01-01	3.000
brokerage	Stock futures	both	2012-01-01	1.000
GST on brokerage + exchange + SEBI				18%
DP charge per scrip per sell day	Delivery	sell		Rs 15.34

Table 1: Per-side cost schedule; the latest row on or before the trade date applies.

4 Results

4.1 Single signals

Strongest neutralized signal: not run, mean rank IC not run (t_{NW} = not run, $h = 5$). Signal tests passing Holm and $t \geq 3$: not run. figure not generated (run `qr figures`)

4.2 Combination

Headline combiner: not run (rank IC not run; equal weight not run; paired difference t = not run).

4.3 Development portfolios

	P1 gross	P1 net	P2 gross	P2 net
Information ratio / Sharpe	not run	not run	not run	not run
CAGR (net)		not run		not run
Max drawdown (net)		not run		not run
One-way turnover per year		not run		not run
Cost drag, bp per year		not run		not run

Table 2: Development window 2017-01-02 to 2024-10-31 (walk-forward). P1 benchmark: Nifty 500 price index. Risk-free: constant 6.5% fallback.

4.4 The holdout read

figure not generated (run `qr figures`)

4.5 Capacity

P1 capacity: Rs not run crore; break-even cost multiple not run. figure not generated (run `qr figures`)

	P1 gross	P1 net	P2 gross	P2 net
Information ratio / Sharpe	not run	not run	not run	not run
95% CI (net)		[not run, not run]		[not run, not run]
PSR(0)		not run		not run
Cost drag, bp per year		not run		not run
STT share of drag, %		not run		not run

Table 3: Locked out-of-sample window 2024-11-01 to 2026-10-30, read once.

5 Robustness

Cost sensitivities (brokerage 0 and 5 bp, impact $Y \in \{0.5, 2\}$, costs doubled) and a 30% delisting haircut are written to `sensitivities_p*.csv` in the portfolio run; they are not new alpha trials and are not in the ledger. Further checks listed in the PRD (no sector neutralization, monthly rebalance, by-year results) are reported only if run.

6 Limitations

- Sector labels are a 2026 snapshot, not point-in-time; names delisted earlier are `Unclassified`.
- No shares-outstanding data: size is proxied by traded value; share turnover is not tested.
- No corporate-action export: splits and bonuses are inferred by ratio matching with F&O lot/open-interest confirmation; dividends are missing (price returns; benchmark is the price index); rights issues are captured only when their ratio is near a common one; an F&O demerger without an open-interest jump would be kept as a fake move. Counts and the 50 most liquid inferred actions are in the data-quality report.
- Protocol incident (disclosed): during a plumbing rehearsal on partial development data a label shuffle silently failed, and the researcher saw the combiner comparison, one signal's IC and the turnover level before the official run. Its 33 trials are in the ledger and in N_{all}; no pre-registered choice was changed (`DECISIONS.md` D-028).
- True delisting returns are unknown; a held name carries flat for up to 10 sessions and is then written off at a 0% or 30% haircut.
- Statutory rates were spot-checked from public sources; the 2024-10-01 futures STT date and the pre-2016 delivery history are not independently confirmed (no reported number depends on pre-2016 rates). Brokerage is assumed; futures margin is a flat 20%; lot rounding is ignored.
- The impact model is a square-root law with $Y = 1$, not fitted to Indian execution data. P2 uses cash-market ADV and volatility for the futures leg.
- The risk-free rate is a constant 6.5% fallback (no RBI T-bill series supplied).
- The trial count is a lower bound: choosing these eight signals from the literature was itself a selection the ledger cannot see. The paper's own deflated-Sharpe worked example was not reproduced (the PSR example was).
- The researcher lived through 2024–2026 markets; the lock prevents numeric peeking, not general knowledge.
- One market, about eight years of walk-forward plus two of holdout.

7 Conclusion

The conclusions are drawn from the ledger values above, whatever they are: a holdout interval that includes zero or a deflated Sharpe below 0.95 is reported as a null result, with the cost attribution.

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